

Q1 Find the correct sequence of prototype software development model.

- A. Start
- B. Develop Prototype
- C. Deliver to customer
- D. Quick Design
- E. Quick Play

Choose the correct answer from the options given below:

- (A) A, B, C, D, E
- (B) A, E, D, B, C
- (C) A, B, D, E, C
- (D) A, B, E D, C

Q2 Consider following conditional statements if ($m = 20 - 10||n > 10$). The order of execution of the following operations is

- A. ==
- B. -
- C. ||
- D. >

Choose the correct answer from the options given below:

- (A) D, B, A, C
- (B) B, D, C, A
- (C) B, D, A, C
- (D) A, B, D, C

Q3 Consider the following statements regarding multiple access techniques & identify the CORRECT ones.

- A. ALOHA is used for multiple access on a shared medium
- B. TDMA uses WALSH TABLES for time slot allocation
- C. CSMA/CD was invented for wireless networks
- D. CDMA codes are sequence of number generated by orthogonal codes

Choose the correct answer from the options given below:

- (A) A & B only
- (B) B & C only
- (C) C & D only
- (D) A & D only

Q4 Which of the following are key features of Mobile Adhoc Networks (MANETs)

- A. Self - Organizing and decentralized.
- B. High Mobility of nodes.

- C. Requires fixed Infrastructure for operation.
- D. Network Topology is dynamic and constantly changing.
- E. It requires a central controller to manage network traffic.

Choose the correct answer from the options given below:

- (A) A, B & E only
- (B) B, C, D & E only
- (C) A, B & D only
- (D) A, B & C only

Q5 Which of the following is not true about Global Variables:

- (A) The values of the Global variables which are sent to the called function may be changed inadvertently by the called function.
- (B) Functions are supposed to be independent and isolated modules. This character is lost, if they use global variables.
- (C) It is not immediately apparent to the reader which values are being sent to the called function
- (D) A function that uses global variables does not suffer from reusability.

Q6 In Software Engineering Jackson's Principle based on

- A. Designation
- B. Definitions
- C. Refutable Assertions
- D. Formal Review
- E. Requirement Elicitation

Choose the correct answer from the options given below:

- (A) A, B & C Only
- (B) B, C & D Only
- (C) C, D & E Only
- (D) B, D & C Only

Q7 Which one of the following is not a basic solution of system of linear equation



$$x_1 + 2x_2 + x_3 = 4$$

$$2x_1 + x_2 + 5x_3 = 5$$

(A) $x_1 = -1, x_2 = 2, x_3 = 1$

(B) $x_1 = 2, x_2 = 1$

(C) $x_1 = 5, x_3 = -1$

(D) $x_2 = 5/3, x_3 = 2/3$

Q8 Match the LIST-I with LIST-II

LIST - I (Software Risk)		LIST - II (Uses)	
A.	Predictable Risk	I.	Strategic Risk
B.	Business Risk	II.	Staffing and Organization
C.	Project Risk	III.	Implementation
D.	Past Risk	IV.	Past Project Experience

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II

(B) A-III, B-II, C-I, D-IV

(C) A-I, B-IV, C-III, D-II

(D) A-IV, B-I, C-II, D-III

Q9 Consider a pipeline unit for fixed - point multiplication of 8-bit integers. Arrange the following stages in a correct sequence.

Stage A : Consists of two CSAs and it merges four numbers from previous stage.

Stage B : is a CPA, which adds up the two numbers

Stage C : is made upto two level of four CSAs.

Stage D : Generates eight partial products.

Choose the correct answer from the options given below:

(A) C, A, B, D

(B) D, C, A, B

(C) B, D, C, A

(D) A, C, D, B

Q10 What will be the right sequence of the phases of database design-

A. Physical Design

B. Conceptual Design

C. Logical Design

D. Requirement collection and analysis

Choose the correct answer from the options given below:

(A) D, A, B, C

(B) D, B, C, A

(C) D, B, A, C

(D) D, A, C, B

Q11 An alternate key in database table is also called

(A) Primary key

(B) Candidate key

(C) Super key

(D) Foreign key

Q12 Which are often needed to evaluate the cache performance?

(A) Hit Ratio

(B) Latency

(C) Cache Traces

(D) Transfer Rate

Q13 Which one of the following is the correct estimation model based on software equation?

(A) $E = [LOC \times B^{0.222/P}]^2 \times (1/t^4)$

(B) $E = [LOC \times B^{0.333/P}]^3 \times (1/t^4)$

(C) $E = [LOC \times B^{0.444/P}]^4 \times (1/t^3)$

(D) $E = [LOC \times B^{0.111/P}]^6 \times (1/t^3)$

Q14 Match the LIST-I with LIST-II: Match the logical equivalence propositions

LIST - I		LIST - II	
A.	$p \rightarrow q$	I.	$(p \wedge q)$ $\vee (\neg p \wedge \neg q)$
B.	$\neg(p \vee (\neg p \wedge q))$	II.	$\neg p \vee q$
C.	$p \leftrightarrow q$	III.	$\neg(p \vee q)$
D.	$\neg(p \leftrightarrow q)$	IV.	$\neg p \leftrightarrow q$

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV

(B) A-I, B-II, C-III, D-IV

(C) A-II, B-III, C-I, D-IV

(D) A-II, B-III, C-IV, D-I

Q15 Match the LIST-I with LIST-II

LIST - I	LIST - II
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A.	Decision Tree	I.	Delta Learning Rule
B.	Supervised Learning	II.	Self Organizing Map
C.	Artificial Neural Network	III.	C4.5 Algorithm
D.	Instance Base Learning	IV.	Non-linear Regression Algorithm

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-III, D-IV
 (B) A-II, B-III, C-IV, D-I
 (C) A-III, B-IV, C-I, D-II
 (D) A-IV, B-I, C-II, D-III

Q16 Consider a noiseless channel with a bandwidth of 5000 Hz transmitting a signal with two signal levels. The maximum bit rate is

- (A) 2500 bps (B) 10000 bps
 (C) 5000 bps (D) 20000 bps

Q17 The hamming distance between 10101 and 11110 is

- (A) 2 (B) 3
 (C) 4 (D) 5

Q18 Arrange the following in the increasing order with respect to access time.

- A. Solid State Drive
 B. Optical Disks
 C. DRAMs
 D. SRAMs
 E. Registers

Choose the correct answer from the options given below:

- (A) D, C, A, B, E (B) B, D, C, A, E
 (C) A, E, B, D, C (D) E, D, C, A, B

Q19 Which of the following is TRUE about the Pumping Lemma for regular language?

- (A)

It is a property used to prove that languages are not regular

- (B) It applies only to infinite regular languages
 (C) It applies to all context - free languages
 (D) It applies to all recursively enumerable languages

Q20 Which of the following is the simplified form of the given function.

$$f(A, B, C) = \sum m(0, 1, 3, 7)$$

- (A) $A'B'C' + A'B'C + A'BC + ABC$
 (B) $A'B' + A'BC + ABC$
 (C) $A'B' + BC$
 (D) $A'B' + BC'$

Q21 Arrange the 2-D viewing transformation pipeline.

- A. Convert world-coordinates to viewing coordinates
 B. Map viewing coordinates to normalized viewing coordinates using window - viewpoint specifications
 C. Construct world - coordinates scene using modeling - coordinates transformations
 D. Map normalized viewpoint to device coordinates

Choose the correct answer from the options given below:

- (A) D, C, B, A (B) D, C, A, B
 (C) C, A, B, D (D) B, C, A, D

Q22 The probability that A hits a target is $\frac{1}{4}$, and the probability that B hits the target is $\frac{2}{5}$. Both shoot at the target, what is the probability that at least one of them hits the target, i.e., that A or B (or both) hit the target?

- (A) $\frac{3}{5}$ (B) $\frac{11}{9}$
 (C) $\frac{2}{20}$ (D) $\frac{11}{20}$

Q23 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Internet message Access (IMAP)	I.	A protocol that allows email clients to retrieve



		emails from a mail server
B.	Address Resolution Protocol (ARP)	II. Used for remove login to another computer over a network
C.	Post Office Protocol (POP)	III. A protocol that maps IP addresses to MAC addresses on a local area network
D.	TELNET	IV. A protocol that allows accessing email on the server without downloading them

Choose the correct answer from the options given below:

- (A) A-II, B-I, C-III, D-IV
- (B) A-III, B-IV, C-I, D-II
- (C) A-IV, B-III, C-I, D-II
- (D) A-IV, B-III, C-II, D-I

Q24 Which table for a dynamic pipeline become more interesting when a nonlinear pattern is follows?

- (A) Reservation Table
- (B) Confusion Table
- (C) Inverted Table
- (D) Greedy Table

Q25 In concurrency control, phantom problem may occur, when-

- (A) records are inserted
- (B) records are deleted
- (C) records are modified
- (D) records are indexed

Q26 Choose the correct statement(s)

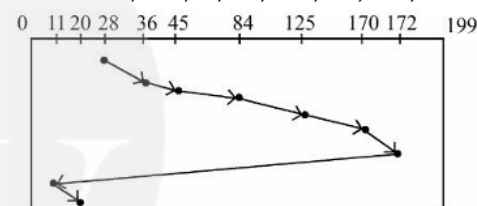
- A. A problem which is NP-Complete will have the property that it can be solved in polynomial time iff all other NP-complete problems can also be solved in polynomial time.
- B. All NP-complete problem are NP-hard problems.
- C. If an NP-hard problem can be solved in polynomial time, then all NP-complete problem can be solved in polynomial time
- D. All NP-hard-problems are not NP-complete.

Choose the correct answer from the options given below:

- (A) A, C only
- (B) B, D only
- (C) A, B, C only
- (D) A, B, C, D

Q27 Consider the following Graph of disk scheduling and identify the disk scheduling algorithm representing by this graph.

Queue 84, 125, 11, 36, 170, 20, 172, 45



- (A) SSTF
- (B) SCAN
- (C) C-SCAN
- (D) C-LOOK

Q28 Software Testability is simply how easily a software program can be tested. Which one characteristics does not lead to testable software?

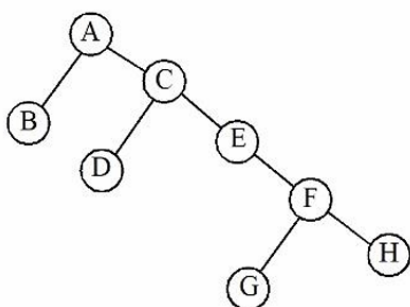
- (A) Observability
- (B) Controllability
- (C) Repairability
- (D) Understandability

Q29 The network using CSMA/CD has a bandwidth of 20 mbps . If the maximum propagation time is 25 μ s (microsecond), what is the minimum size of the frame?

- (A) 500 bits
- (B) 1000 bits
- (C) 1500 bits
- (D) 2000 bits

Q30





Considering above binary tree, what will be the inorder traversal

- (A) BADCEGFH (B) GHFEDCBA
(C) BACDEGFH (D) GHFDEBCA

Q31 Regarding the code optimization, choose the correct sequence

- A. Algebraic Simplification
B. Use of machine idioms
C. Redundant - instruction elimination
D. Flow of control optimization
E. Improved target code

Choose the correct answer from the options given below:

- (A) A, C, B, D, E (B) B, C, D, A, E
(C) C, D, A, B, E (D) D, B, A, C, E

Q32 Sequence the following stages of the sliding window protocol for flow control.

- A. Sender move the window to the next set of frames.
B. Sender sends a window of frames.
C. Receiver acknowledge the received frames.
D. If acknowledgment is not received, Sender retransmits the frame.

Choose the correct answer from the options given below:

- (A) B, A, C, D (B) B, C, D, A
(C) B, D, C, A (D) B, C, A, D

Q33 Which of the following uses only increment operations for adding and removing element at either end?

- (A) Queues
(B) Stacks
(C) Priority Queues
(D) Deques

Q34

Which of the following/s is/are FALSE statement?

- A. An all key relation is always in BCNF since it has no FDs.
B. A relation that is not in 4NF due to nontrivial MVD must be decomposed to convert it into a set of relations in 4NF.
C. The decomposition removes the redundancy by the MVD.
D. 3NF is stronger than BCNF.

Choose the correct answer from the options given below:

- (A) A only (B) A, B only
(C) D only (D) C only

Q35 Which of the following command can be used to modify data in an SQL Table?

- A. INSERT
B. DELETE
C. UPDATE
D. SELECT

Choose the correct answer from the options given below:

- (A) A, B, D only (B) A, C, D only
(C) B, C, D only (D) A, B, C only

Q36 The entire set of parameters, including return address that is stored for a procedure invocation is referred to

- (A) stack frame (B) stack base
(C) stack limit (D) stack record

Q37 Which of the following represents the output of the transition function(δ)

$$\delta(q_0, a) = (q_1, x, R)$$

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, y) = (q_1, y, R)$$

$$\delta(q_1, b) = (q_2, y, L)$$

$$\delta(q_2, y) = (q_2, y, L)$$

$$\delta(q_2, a) = (q_2, a, L)$$

$$\delta(q_2, x) = (q_0, x, R)$$

$$\delta(q_0, y) = (q_3, y, R)$$

$$\delta(q_3, y) = (q_3, y, R)$$

$$\delta(q_3, \square) = (q_f, \square, R)$$

- (A) L (B) L
 $= \{a^n b^n \mid n \geq 0\}$ $= \{a^n b^n \mid n \geq 1\}$



$$(C) L = \{a^n b^n \mid n > 0\} \quad (D) L = \{a^n b^n \mid n > 1\}$$

Q38 Correct the order of instruction cycle:

- A. Read the effective address
- B. Fetch the information
- C. Execute the instruction
- D. Decode the instruction

Choose the correct answer from the options given below:

- (A) A, B, C, D
- (B) B, D, A, C
- (C) B, A, D, C
- (D) A, B, D, C

Q39 Identify the correct statements from the following regarding spiral model.

- A. Spiral model is an incremental process model.
- B. Spiral model is a risk - driven process mode.
- C. Spiral model is an acyclic balancing approach.
- D. Spiral model involves a set of anchor point milestone.

Choose the correct answer from the options given below:

- (A) A & B Only
- (B) B & C Only
- (C) A & D Only
- (D) B & D Only

Q40 If x and y are elements in a group G and if $x^5 = y^3 = e$, where e is the identity of G , then the inverse of $x^2 y x^4 y^2$ must be

- (A) $y^2 x y^2 x^4$
- (B) $yx y^2 x^3$
- (C) $yx^6 y^6 x^3$
- (D) $x^4 y^2 x^2 y$

Q41 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	RAID level 1	I.	bit-interleaved parity organization
B.	RAID level 2	II.	disk mirroring
C.	RAID level 3	III.	block-interleaved parity organization

D.	RAID level 4	IV.	ECC organization
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Choose the correct answer from the options given below:

- (A) A-II, B-I, C-IV, D-III
- (B) A-II, B-IV, C-I, D-III
- (C) A-III, B-IV, C-I, D-II
- (D) A-IV, B-III, C-II, D-I

Q42 Which of the following are correct for the neural network?

- A. The training time depends upon the size of network.
- B. Neural network can be simulated on the conventional computer.
- C. Neural network mimic the same way as that of the humans brain.
- D. A neural network include feedback.

Choose the correct answer from the options given below:

- (A) A and B only
- (B) A, C and D only
- (C) A, B and C only
- (D) A and C only

Q43 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Conflict Resolution	I.	Three-Address Code
B.	Common Subsequence	II.	Precedence
C.	Quadruples	III.	Syntax Directed Translations
D.	L-Attributed definitions	IV.	Direct Acyclic Graph (DAG)

Choose the correct answer from the options given below:

- (A) A-III, B-II, C-IV, D-I
- (B) A-II, B-IV, C-I, D-III



- (C) A-II, B-III, C-I, D-IV
(D) A-IV, B-I, C-II, D-III

Q44 What is the correct sequence of steps used by knowledge base designing?

- A. Ask questions about the intended interpretation.
B. Chose task domain or world to represent.
C. Select atoms to represent propositions of Interest.
D. Tells the system propositions that are true in the intended interpretation/axiomatizing the domain.

Choose the correct answer from the options given below:

- (A) A, B, C, D (B) B, C, D, A
(C) C, D, A, B (D) D, A, B, C

Q45 The mathematical notation to describe logical entailment of a sentence " α entails another sentence β " is

- (A) $\alpha \models \beta$
(B) $\alpha \subseteq \beta$
(C) $\beta \models \alpha$
(D) $\beta \subseteq \alpha$

Q46 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Breadth First Search	I.	LISP
B.	Depth First Search	II.	Syntax tree
C.	Prefix	III.	Stack
D.	Infix	IV.	Queue

Choose the correct answer from the options given below:

- (A) A-III, B-IV, C-I, D-II
(B) A-III, B-IV, C-II, D-I
(C) A-IV, B-II, C-III, D-I
(D) A-IV, B-III, C-I, D-II

Q47 If the virtual memory size is 32 MB and a physical memory size is 4 MB with a page size of 2 KB ,

Calculate the number of frames available in physical memory

- (A) 1024 (B) 2048
(C) 3072 (D) 4096

Q48 Collaborative document inspection technique used for?

- (A) Requirement Validation Process
(B) Requirement Verification Process
(C) Integration Testing
(D) Blackbox Testing

Q49 The kind of symbols for basic syntactic elements of first-order logic are

- A. Constant
B. Domain
C. Predicate
D. Temporal
E. Function

Choose the correct answer from the options given below:

- (A) B, D only (B) A, B, C only
(C) A, C, E only (D) C, D only

Q50 Consider the following statements

`int x = 10, y = 15;`

`x = ((x == y)?(y + x) : (y - x));`

What will be the value of x after executing these statements?

- (A) 5 (B) 25
(C) 15 (D) 30

Q51 The major adverse side effects of scan conversion are ;-

- A. Staircase appearance
B. Unequal brightness of slanted lines
C. Picket fence problem
D. Rasterization
E. Pre-filtering and Post-filtering

Choose the correct answer from the options given below:

- (A) A & B only
(B) A, B & C only
(C) B, C & D only
(D) C, D & E only

Q52



The primary function of firewall in network security is

- (A) To monitor network traffic and detect anomalies.
- (B) To create virtual private networks for secure communication.
- (C) To filter incoming and outgoing network traffic based on security rules
- (D) To encrypt data before Transmission.

Q53 Match the LIST-I with LIST-II

LIST - I (Functions)		LIST - II (Uses of functions)	
A.	fseek()	I.	Gives the current position in the file
B.	ftell()	II.	Sets the position to the beginning of the file
C.	feof()	III.	Sets the position to a desired point in the file
D.	rewind()	IV.	Used to test for an end of file condition.

Choose the correct answer from the options given below:

- (A) A-II, B-III, C-I, D-IV
- (B) A-III, B-I, C-IV, D-II
- (C) A-III, B-I, C-II, D-IV
- (D) A-II, B-IV, C-III, D-I

Q54 Which of the following table contains the primary information in the data warehouse?

- (A) Dimension table
- (B) Fact table
- (C) Lookup table

(D) Primary table

Q55 INSERT command is used to

- A. add a single tuple to a relation
- B. add multiple tuples to a relation
- C. add values to specific attributes
- D. insert new table

Choose the correct answer from the options given below:

- (A) A only
- (B) A, B only
- (C) A, B, C only
- (D) B, C, D only

Q56 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Type - 0 Grammar	I.	Linear Grammar
B.	Type - 1 Grammar	II.	GNF
C.	Type - 2 Grammar	III.	$x \rightarrow y, x < y ; x, y \in \{V \cup T\}^*$
D.	Type - 3 Grammar	IV.	Recursively Enumerable

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-III, D-IV
- (B) A-II, B-I, C-III, D-IV
- (C) A-III, B-II, C-IV, D-I
- (D) A-IV, B-III, C-I, D-II

Q57 Which one of the following is not the part of requirement management?

- (A) Features Traceability Table
- (B) Dependency Traceability Table
- (C) Interface Traceability Table
- (D) Page Traceability Table

Q58 Arrange the following graph on the basis of number of edges in increasing order [for $n > 3$]

- A. K_n (Complete Graph)
- B. C_n (Cycle graph)
- C. W_n (Wheel graph)



D. $K_{n,n}$ (Complete Bipartite Graphs)

E. Q_n (n-cubes graph)

Choose the correct answer from the options given below:

- (A) A, B, C, D, E (B) B, C, A, D, E
(C) A, B, C, E, D (D) E, D, C, A, B

Q59 Which of the following is the complement of the boolean function. $A'B + CD' + A'B + CD'$

(A) $A'B + CD'$ (B) $(A' + B)(C + D')$
(C) $(A + B')$ (D) $AB' + CD'$
(E) $(C' + D)$

Q60 Which of the following is the correct sequence with regard to service routine?

- A. Save contents of processor registers
- B. Turn the interrupt facility on
- C. Service the device whose flag is set
- D. Check which flag is set
- E. Restore contents of processor registers

Choose the correct answer from the options given below:

- (A) A, B, C, D, E (B) D, B, C, A, E
(C) D, C, B, E, A (D) A, D, C, E, B

Q61 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Hadoop Distribution File System	I.	Trust
B.	Veracity	II.	Facebook
C.	Apache Pig	III.	Name-Node
D.	Apache Hive	IV.	Yahoo

Choose the correct answer from the options given below:

- (A) A-IV, B-III, C-I, D-II
(B) A-III, B-I, C-II, D-IV
(C) A-III, B-I, C-IV, D-II
(D) A-II, B-III, C-I, D-IV

Q62 A linear programming problem (LPP) is as follows:
Min $z = 30x - 18y$, subject to the constraints;

$3x + 4y \leq 60, 5x - 3y \geq 20$ and $x, y \geq 0$. In

this feasible region, the solution of LPP is/are

- A. (4, 0)
B. (2, 0)
C. (7, 5)
D. (0, 15)
E. (8, 5)

Choose the correct answer from the options given below:

- (A) A and C only (B) B only
(C) E only (D) D only

Q63 Which of the following is not a divide and conquer method

- (A) Binary Search (B) Merge Sort
(C) Quick Sort (D) Heap Sort

Q64 Which of the following statement is not True?

- (A) A data member of a class can be declared as static and is normally used to maintain values common to the entire class.
- (B) A friend function can be invoked like a normal function without the help of any object.
- (C) The scope resolution operator (::) can be overloaded like normal operators
- (D) Multiple inheritance may lead to duplication of inherited members from a "Grandparent" base class. This may be avoided by making the common base class a virtual base class.

Q65 The degree of a relation in database is

- (A) the number of tuples in the relation
(B) the number of attribute in the relation
(C) the number of values in domian
(D) the number of keys in a relation

Q66 Which of the following CFG(s) is/are in chomsky Normal form (All capital letters are variables & lower case are terminals)

A. $S \rightarrow ABC/AB$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$C \rightarrow d$$

B. $X \rightarrow RT/TR$

$$T \rightarrow t$$

$$R \rightarrow XT/r$$



$$C. P \rightarrow qP/sQ$$

$$Q \rightarrow r/s$$

$$D. M \rightarrow MN/MP$$

$$N \rightarrow nm/n$$

$$P \rightarrow p$$

Choose the correct answer from the options given below:

- (A) A & B only (B) B only
(C) C only (D) B & D only

Q67 Arrange the following steps of the Inorder Traversal of Binary Tree in the correct order.

- A. Visit the Left subtree
B. Visit the Root node
C. Visit the Right subtree
D. Start traversing by visiting the nodes in rooted tree
E. Repeat the above three steps.

Choose the correct answer from the options given below:

- (A) A, B, C, D, E (B) E, A, B, C, D
(C) D, A, B, C, E (D) D, A, E, B, C

Q68 Match the LIST-I with LIST-II

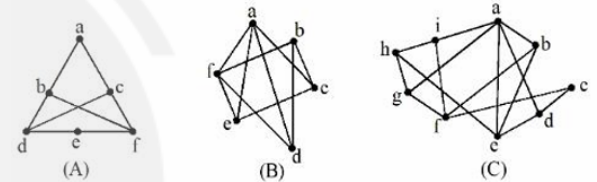
LIST - I		LIST - II	
A.	Service Level Agreement	I.	The technology that connects and enables communication between physical devices over the Internet.
B.	Virtualization	II.	A develop environment provides over the Internet to create and deploy applications.
C.	Internet of Things	III.	Contracts that specify performance

		standards and services availability.
D.	Platform as a service	IV. A method allowing multiple operating system to run on a single physical server.

Choose the correct answer from the options given below:

- (A) A-III, B-II, C-I, D-IV
(B) A-III, B-IV, C-I, D-II
(C) A-III, B-I, C-IV, D-I
(D) A-I, B-IV, C-III, D-II

Q69 Which of the following Graph is/are planer?



Choose the most appropriate answer from the options given below:

- (A) A and C only (B) B only
(C) A only (D) A and B only

Q70 Choose the correct statements from the following

- A. In connection oriented service, the destination address is to be specified only during the setup.
B. Packet sequencing is not guaranteed in connection oriented service
C. Flooding is method in which every incoming packet is sent out on every outgoing line except the one by which it arrived.

- D. Presentation layer performs detection and recovery from errors in the transmitted data

Choose the correct answer from the options given below:

- (A) A, B & C only
(B) B, C & D only
(C) A & C only
(D) A, C & D only



- Q71** Which of the following algorithms are based on the Breadth First Search (BFS) ?
- Prim's algorithms
 - Kruskal algorithms
 - Dijkstra algorithms
 - Greedy algorithms
 - Dynamic Programming
- Choose the correct answer from the options given below:
- A & B only
 - A, C & D only
 - D & E only
 - A & C only
- Q72** Which of the following is the solution of the following recurrence relation
 $T(n) = T(2n/3) + 1$?
- $\theta(n^2)$
 - $\theta(\log n)$
 - $\theta(n \log n)$
 - $\theta(n^{3/2})$
- Q73** Which of the following is not a component of the classic Planning Definition?
- Init
 - Domain
 - Action
 - Goal
- Q74** y_{acc} is which of the following parsers?
- Predictive Parser
 - SLR Parser
 - CLR Parser
 - LALR Parser
- Q75** Definition's of _____ organized into following four categories namely, Thinking Humanly, Thinking Rationally, Acting Humanly, Acting Rationally
- Machine Learning
 - Deep Learning
 - Artificial Intelligence
 - Neural Network
- Q76** Which of the following does not interrupt a running process?
- Device
 - Timer
 - Scheduler
 - Power Failure
- Q77** Given the following processes with their times and priority (lower number indicates higher

priority)

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	4
P4	1	5
P5	5	2

Which of the following is a value of average waiting time using priority scheduling algorithm?

- 9.2
- 6.8
- 7.6
- 8.2

- Q78** Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Project Join Normal Form	I.	Concurrency
B.	The lost update problem	II.	Fifth Normal Form
C.	Discretionary Access Control	III.	SQL
D.	Data Manipulation Language	IV.	Security

Choose the correct answer from the options given below:

- A-II, B-I, C-IV, D-III
- A-I, B-II, C-IV, D-III
- A-I, B-II, C-III, D-IV
- A-IV, B-II, C-III, D-I

- Q79** Match the LIST-I with LIST-II

LIST - I (Boolean Algebra Law)		LIST - II (Axioms)	
A.	Absorption Law	I.	$a + 1 = 1$
B.	Bounded Law	II.	$a + 0 = a$



C.	Identity Law	III.	$a * (b + c) = (a * b) + (a * c)$
D.	Distributive Law	IV.	$a + (a * b) = a$

Choose the correct answer from the options given below:

- (A) A-IV, B-I, C-II, D-III
 (B) A-IV, B-III, C-I, D-II
 (C) A-III, B-IV, C-II, D-I
 (D) A-II, B-III, C-IV, D-I

Q80 The light given off by the phosphor during exposure to the electron beam is known as

- (A) Fluorescence
 (B) Phosphorescence
 (C) Persistence
 (D) Retracing

Q81 Which of the following description(s) is/are true:

- A. Red Black Tree - Guarantees worst case of $O(\log n)$ time for search, insert & delete.
 B. Trie - Used for efficient prefix - based searches.
 C. AVL Tree - self balancing binary search tree with stricter balance criteria.
 D. B- Tree - Allows efficient search, insert, delete operations in disk based system.

Choose the correct answer from the options given below:

- (A) A & B only
 (B) C & D only
 (C) B only
 (D) A, B, C & D

Q82 Match the LIST-I with LIST-II

LIST - I	LIST - II
----------	-----------

A. $(100110011)_2$	I.	Error detection code
B. Complements	II.	CISC
C. Parity Bits	III.	Negative number
D. Variable length instruction	IV.	$(133)_{16}$

Choose the correct answer from the options given below:

- (A) A-IV, B-I, C-III, D-II
 (B) A-IV, B-III, C-I, D-II
 (C) A-IV, B-II, C-III, D-I
 (D) A-I, B-IV, C-II, D-III

Q83 Which of the following are not data manipulation instructions?

- A. Call
 B. Loud
 C. And
 D. Increment
 E. Shift

Choose the correct answer from the options given below:

- (A) C & E only
 (B) A & B only
 (C) D & E only
 (D) A & C only

Q84 Which of the following do not represent memory reference instruction.

- A.

1	opcode	
---	--------	--

 mode = 1, opcode = 010
 B.

0	opcode	
---	--------	--

 mode = 0, opcode = 111
 C.

1	opcode	
---	--------	--

 mode = 1, opcode = 111
 D.

1	opcode	
---	--------	--

 mode = 1, opcode = 110
 E.

1	opcode	
---	--------	--

 mode = 1, opcode = 001

Choose the correct answer from the options given below:

- (A) B & C only
 (B) B & D only
 (C) A & D only
 (D) A & E only

Q85 Which one of the following is not related to the feed forward networks on the Backpropagation Algorithm

- (A) Boolean function
 (B) Continuous function



- (C) Arbitrary function
(D) Greedy function

Q86 Match the LIST-I with LIST-II

LIST - I		LIST - II	
A.	Insertion Sort	I.	$O(\log n)$
B.	Binary Search	II.	$O(n^2)$
C.	Quick Sort	III.	$O(n)$
D.	Selection Sort	IV.	$O(n \log n)$

Choose the correct answer from the options given below:

- (A) A-III, B-I, C-IV, D-II
(B) A-II, B-III, C-I, D-IV
(C) A-I, B-II, C-IV, D-III
(D) A-II, B-III, C-IV, D-I

Q87 As per the Software Engineering Institute (SEI), the correct sequence is

- A. Initial
B. Repeatable
C. Managed
D. Optimizing
E. Defined

Choose the correct answer from the options given below:

- (A) A, B, C, D, E (B) A, B, E, C, D
(C) A, E, C, D, B (D) A, E, B, C, D

Q88 Arrange the following steps of the DHCP (Dynamic Host Control Protocol) process in the correct - sequence.

- A. DHC PACK
B. DHCP DISCOVER
C. DHCP OFFER
D. DHCP REQUEST
E. Client receives IP configuration

Choose the correct answer from the options given below:

- (A) A, B, C, D, E (B) B, C, D, A, E
(C) C, B, A, E, D (D) D, A, E, C, B

Q89 Which of the following statements are true about the sets.

- A. $0 \in \emptyset$
B. $\emptyset \in \{0\}$
C. $\emptyset \in \{\emptyset\}$
D. $\{\emptyset\} \in \{\emptyset\}$
E. $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$

Choose the correct answer from the options given below:

- (A) A, B, C, D and E
(B) A, B, C and E only
(C) A, C only
(D) C and E only

Q90 The characteristics of modern programming are :

- A. Each module should do only one thing.
B. Communication between modules is allowed only by a calling module.
C. Communication can take place directly between modules that do not have calling - called relationship.
D. All modules are designed as single-entry, single exit systems using control structure.

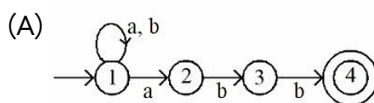
Choose the correct answer from the options given below:

- (A) A, B & C only
(B) A, B & D only
(C) A, C & D only
(D) C & D only

Directions (91-95) Read the following passage and answer the given questions.

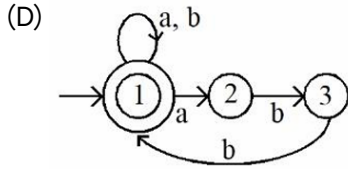
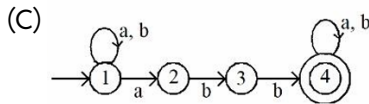
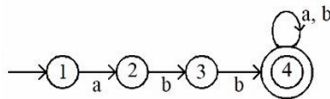
A machine is represented by states Q , input alphabet Σ , transition function $\hat{o}$. Initial state q_0 and final state F . The machine accepts all the strings over $\Sigma = \{a, b\}$, which start and ended with any combination of all alphabet and abb works/lies as substring in all the strings to be accepted.

Q91 For the above mentioned passage which of the following is correct?



(B)





Q92 For the above specified passage, which of the following represent the grammar for the language accepted the machine?

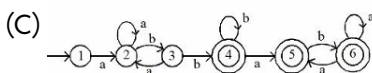
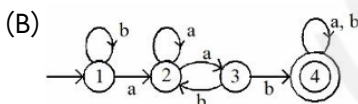
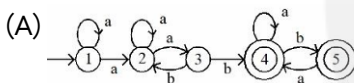
(A) $S \rightarrow AabbB, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid \epsilon$

(B) $S \rightarrow abbA, A \rightarrow aA \mid \epsilon \mid bA$

(C) $S \rightarrow AabbA, A \rightarrow aA \mid bA \mid \epsilon$

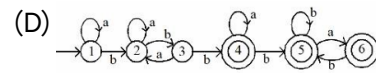
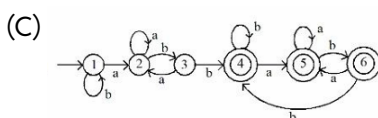
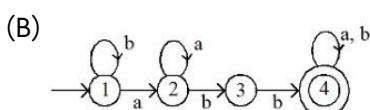
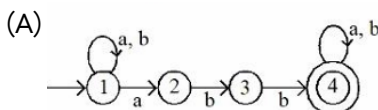
(D) $S \rightarrow Aabb, A \rightarrow aA \mid bA \mid \epsilon$

Q93 Which of the following represented the minimum state DFA for the above specified passage?



(D) All are NFA

Q94 For the above specified passage, which of the following is DFA for the language represented/accepted by machine?



Q95 For the above specified passage, which of the following represents the regular expression?

(A) $(a + b)^*aab$

(B) $aba(a + b)^*$

(C) $b(a + b)^*b(a + b)^*a(a + b)^*$

(D) $(a + b)^*abb(a + b)^*$

Directions (96-100) Read the following passage and answer the given questions.

Consider the following set of processes with the arrival time and length of CPU Burst time given in milli-seconds (ms):

Process	Arrival Time	Burst Time
P1	0	5
P2	2	3
P3	2	2
P4	5	3
P5	6	1

Q96 Calculate the average response time for these processes with nonpreemptive priority Scheduling algorithm? Priority of the process are as follows and consider that all the processes came at zero (0) time. (Note low numbers represent high priority).

Process	Burst Time	Priority
P1	5	3
P2	3	1
P3	2	5
P4	3	4
P5	1	2

(A) 5.6 ms

(B) 5.2 ms

(C) 4.7 ms

(D) 3.8 ms

Q97



Find the average weighted turnaround time for these process using Highest Response Ratio Next (HRRN) Algorithm?

- (A) 3.2 ms (B) 2.834 ms
(C) 1.632 ms (D) 6.721 ms

Q98 What is the average turnaround time for these Processes with First Come First Serve (FCFS) scheduling Algorithm?

- (A) 5.3 ms (B) 6.4 ms
(C) 7.0 ms (D) 8.2 ms

Q99 What is the average waiting time for these processes with non-preemptive Shortest

Job First (SJF) scheduling Algorithm?

- (A) 3.0 ms (B) 2.1 ms
(C) 4.6 ms (D) 3.2 ms

Q100 When we apply Shortest Job First (SJF) scheduling Policy/algorithm, Process (P2) faces partial starvation and waits for the longer time to execute. Which one of the following algorithms addresses this problem?

- (A) Round Robin
(B) Priority
(C) Highest Response ratio Next
(D) Least Completed next



Answer Key

Q1 (B)
Q2 (C)
Q3 (D)
Q4 (C)
Q5 (D)
Q6 (A)
Q7 (A)
Q8 (D)
Q9 (B)
Q10 (B)
Q11 (B)
Q12 (A)
Q13 (B)
Q14 (C)
Q15 (C)
Q16 (B)
Q17 (B)
Q18 (D)
Q19 (A)
Q20 (C)
Q21 (C)
Q22 (D)
Q23 (C)
Q24 (A)
Q25 (A)
Q26 (D)
Q27 (D)
Q28 (C)
Q29 (B)
Q30 (A)
Q31 (C)

Q32 (D)
Q33 (D)
Q34 (C)
Q35 (D)
Q36 (A)
Q37 (B)
Q38 (B)
Q39 (D)
Q40 (B)
Q41 (B)
Q42 (B)
Q43 (B)
Q44 (B)
Q45 (A)
Q46 (D)
Q47 (B)
Q48 (A)
Q49 (C)
Q50 (A)
Q51 (B)
Q52 (C)
Q53 (B)
Q54 (B)
Q55 (C)
Q56 (D)
Q57 (D)
Q58 (B)
Q59 (C)
Q60 (D)
Q61 (C)
Q62 (A)



Q63 (D)
Q64 (C)
Q65 (B)
Q66 (B)
Q67 (C)
Q68 (B)
Q69 (D)
Q70 (C)
Q71 (D)
Q72 (B)
Q73 (B)
Q74 (D)
Q75 (C)
Q76 (C)
Q77 (D)
Q78 (A)
Q79 (A)
Q80 (A)
Q81 (D)

Q82 (B)
Q83 (B)
Q84 (A)
Q85 (D)
Q86 (A)
Q87 (B)
Q88 (B)
Q89 (D)
Q90 (B)
Q91 (C)
Q92 (C)
Q93 (D)
Q94 (C)
Q95 (D)
Q96 (A)
Q97 (B)
Q98 (C)
Q99 (D)
Q100 (C)



Hints & Solutions

Q1 Text Solution:

The steps of the Prototype Model are:

Start (A): Initiate the process.

Quick Play (E): Identify the basic requirements and prepare for prototyping.

Quick Design (D): Create a quick design focusing on the functionality of the prototype.

Develop Prototype (B): Build the prototype based on the quick design.

Deliver to Customer (C): Present the prototype to the customer for feedback and further refinement.

This sequence aligns with the iterative nature of the prototype model, focusing on fast design and development cycles for early customer feedback. The correct sequence of the Prototype Software Development Model is: Option B: A, E, D, B, C

Q2 Text Solution:

if ($m == 20 - 10 \parallel n > 10$)

Operator Precedence & Associativity-

The execution order of operations is:

B. Subtraction (-) $\rightarrow 20 - 10$

D. Greater than (>) $\rightarrow n > 10$

A. Equality (==) $\rightarrow m == 10$

C. Logical OR ($\parallel$)

Correct Order is : **Option C:** B,D, A,C

Q3 Text Solution:

Statement A: "ALOHA is used for multiple access on a shared medium"

Correct: ALOHA is a random access protocol used for multiple access in a shared medium (such as satellite and wireless communication). It allows multiple devices to transmit data over the same channel.

Statement B: "TDMA uses WALSH TABLES for time slot allocation"

Incorrect: TDMA (Time Division Multiple Access) allocates time slots to users, but it does not use Walsh tables.

Walsh tables are used in CDMA (Code Division

Multiple Access) for orthogonal code generation, not TDMA.

Statement C: "CSMA/CD was invented for wireless networks"

Incorrect: CSMA/CD (Carrier Sense Multiple Access with Collision Detection) was designed for wired Ethernet networks, not wireless. Wireless networks use CSMA/CA (Collision Avoidance) instead of CSMA/CD because detecting collisions is difficult in a wireless medium.

Statement D: "CDMA codes are sequence of numbers generated by orthogonal codes"

Correct: CDMA (Code Division Multiple Access) uses orthogonal codes, such as Walsh codes, to separate signals.

These codes allow multiple users to transmit over the same frequency simultaneously without interference.

Correct Answer: Option D (A & D only)

Q4 Text Solution:

Key Features of Mobile Adhoc Networks (MANETs):

1. Self-Organizing and Decentralized (A):

MANETs do not rely on centralized infrastructure. Nodes organize and communicate among themselves.

2. High Mobility of Nodes (B):

Nodes in MANETs can move freely, resulting in dynamic and frequently changing network topology.

3. Dynamic Topology (D):

The network's structure changes continuously as nodes join, leave, or move.

Incorrect Options:

C. Requires Fixed Infrastructure (Incorrect):

MANETs are infrastructure-less networks, meaning they do not require fixed infrastructure.

E. Requires a Central Controller (Incorrect):

MANETs are decentralized and do not rely on a central controller for managing network traffic.



Q5 Text Solution:**Option D (Not True):**

Using global variables does affect reusability, as the function's reliance on external global variables makes it harder to reuse independently.

- Functions that depend on global variables are tightly coupled with them, making it difficult to reuse the function in different contexts without bringing the global variable along.
- Functions should ideally be self-contained, taking input via arguments and returning output.

Q6 Text Solution:

Jackson's Principle : Jackson's Principle is a concept used in software engineering to address designation, definitions, and refutable assertions during software design and development. Here's how these elements are relevant:

Designation (A): Refers to assigning clear names or identifiers to components in the system, ensuring clarity and traceability.

Definitions (B): Ensures all components, terms, and relationships are well-defined and unambiguous.

Refutable Assertions (C): Involves making assertions that are testable and refutable, allowing for verification and validation.

Why the Other Options are Incorrect:

D. Formal Review: While important in software engineering, this is not part of Jackson's Principle.

E. Requirement Elicitation: This is related to gathering requirements but not directly linked to Jackson's Principle.

Thus, the key components of Jackson's Principle are Designation, Definitions, and Refutable Assertions, making Option A the correct answer.

Q7 Text Solution:

There are three variables (x_1, x_2, x_3) and two equations. A basic solution is obtained by setting one variable to zero and solving for the other two.

Analyzing Given Options

Option A: $x_1=-1, x_2=2, x_3=1$

- No zero variable is present.
- This **cannot be a basic solution**.

Option B: $x_1=2, x_2=1$ (**implicitly $x_3=0$**)

- Matches the computed basic solution (2,1,0).
- This is a basic solution.

Option C: $x_1=5, x_3=-1$ (**implicitly $x_2=0$**)

- Matches the computed basic solution (5,0,-1).
- This is a basic solution.

Option D: $x_2=5/3, x_3=2/3$ (**implicitly $x_1=0$**)

- Matches the computed basic solution (0,5/3,2/3) (0,35,32).
- This is a basic solution.

Conclusion

- Option A ($x_1=-1, x_2=2, x_3=1$) is not a basic solution because none of the variables are zero.
- The other three options represent valid basic solutions.

Q8 Text Solution:

LIST - I (Software Risk)	LIST - II (Uses)	Explanation
1. Predictable Risk	1. Project Experience	Predictable Risk: This typically refers to risks that



			can be anticipated based on previous experiences. The use related to past experiences is often
			Past Project Experience, which helps in identifying these risks.

B	Bu	I.	Str	Busi
.	sin	tegi	nes	
es	c	Risk	-	
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k			kind	
			of	
			risk	
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Q9 Text Solution:

The steps involved in the fixed-point multiplication process for 8-bit integers and the functions of each stage:

Stage D generates **eight partial products**. In multiplication, partial products are formed by multiplying each bit of one number with all bits of the other number.

Stage C is made up of two levels of **four CSAs (Carry-Save Adders)**, which help in adding the partial products.

Stage A consists of **two CSAs** and merges four numbers from the previous stage. This stage

helps reduce the number of partial products into fewer numbers for addition.

Stage B is a **CPA (Carry Propagate Adder)**, which adds up the two numbers to finalize the multiplication result.

Thus, the correct order is **D, C, A, B**.

Q10 Text Solution:

The correct sequence of phases for database design is as follows:

Requirement Collection and Analysis (D): This is the first step where the needs and requirements of the system are gathered, and the objectives of the database system are defined.

Conceptual Design (B): In this phase, a high-level model of the database is created, which is independent of the database management system (DBMS) and physical aspects. It focuses on the entities, their relationships, and constraints.

Logical Design (C): This phase involves translating the conceptual design into a logical structure that can be implemented in a specific DBMS, such as a relational model.

Physical Design (A): This phase focuses on the physical storage and optimization of the database, deciding how data will be stored and accessed in the DBMS.

The correct sequence is **D, B, C, A**.

Q11 Text Solution:

An **alternate key** in a database table is also called a **Candidate Key**. This is because, in addition to the **primary key**, other keys in the table that can uniquely identify a record are called candidate keys. If any of these candidate keys are not selected as the primary key, they are referred to as alternate keys.

Thus, the correct answer is **B - Candidate key**.

Q12 Text Solution:

The **hit ratio** is one of the key performance indicators used to evaluate the effectiveness of a **cache system**. It is defined as the fraction or percentage of memory accesses that are satisfied by the cache without needing to access the slower main memory.



Formula:

$$= \frac{\text{Hit Ratio}}{\text{Number of Cache Hits}} \\ = \frac{\text{Total Number of Memory Accesses}}{\text{Total Number of Memory Accesses}}$$

Hit ratio is a vital metric for cache evaluation because it directly measures how effectively the cache is reducing the need for slower memory access, which is crucial for system performance.

Q13 Text Solution:

General Software Estimation Model:

The software equation is given by:

$$E = \left[\frac{\text{LOC} \times B^k}{P} \right]^n \times \frac{1}{t^m}$$

Where:

- E: Effort (measured in person-months or person-hours),
- LOC: Lines of Code (size of the software project),
- B: Constant depending on the type of project,
- P: Productivity parameter,
- t: Time (development time in months),
- k, n, and m: Exponents derived from empirical data.

1. The exponent k for B^k is typically **0.333**, which aligns with the software estimation models in empirical studies.

2. The power of n is related to effort estimation; in this equation, it's represented as **3**.

3. The time exponent m is most commonly found as **4**, indicating the inverse proportionality of effort to the fourth power of development time.

Therefore, the correct answer is **Option B**.

$$E = \left[\frac{\text{LOC} \times B^{0.333}}{P} \right]^3 \times \frac{1}{t^4}$$

Q14 Text Solution:

(A) $p \rightarrow q$ (A matches with II)

- Implication ($p \rightarrow q$) is logically equivalent to $\neg p \vee q$.

(B) $\neg(p \vee (\neg p \wedge q))$ (B matches with III)

- Distribute **De Morgan's Law**:

$$\neg p \wedge \neg(\neg p \wedge q)$$

- Since $\neg p \wedge q$ means "q when p is false," negating it gives $\neg q \vee p$.
- So,
 $\neg(p \vee (\neg p \wedge q)) = \neg p \wedge \neg q = \neg(p \vee q)$

(C) $p \leftrightarrow q$ (C matches with I)

- $p \leftrightarrow q$ means both have the same truth value:

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$

(D) $\neg(p \leftrightarrow q)$ (D matches with IV)

- $\neg(p \leftrightarrow q)$ is the negation of equivalence.
- Since $p \leftrightarrow q$ is $(p \wedge q) \vee (\neg p \wedge \neg q)$, its negation is:

$$(p \wedge \neg q) \vee (\neg p \wedge q)$$

- This is logically equivalent to $\neg p \leftrightarrow q$.

Q15 Text Solution:

A. Decision Tree $\rightarrow$ C4.5 Algorithm (III)

- **Reason:** Decision trees are commonly built using algorithms like **C4.5**, which helps in classification tasks.

B. Supervised Learning $\rightarrow$ Non-linear Regression Algorithm (IV)

- **Reason:** Supervised learning includes **regression models**, where **non-linear regression** is used to model complex relationships.

C. Artificial Neural Network (ANN) $\rightarrow$ Delta Learning Rule (I)

- **Reason:** The **Delta Learning Rule** is used for adjusting weights in **Artificial Neural Networks (ANNs)** during training.

D. Instance-Based Learning $\rightarrow$ Self-Organizing Map (II)

- **Reason:** **Instance-based learning** relies on memorizing instances, and **Self-Organizing Maps (SOMs)**, which are used for clustering, fall into this category.



Q16 Text Solution:

For a **noiseless channel**, the maximum bit rate is given by **Nyquist's theorem**:

$$\text{Bit Rate} = 2 \times B \times \log_2(M)$$

Where:

- $B = 5000$ Hz (Bandwidth)
- $M = 2$ (Number of signal levels)
- $\log_2(2) = 1$

$$\text{Bit Rate} = 2 \times 5000 \times 1$$

Bit Rate = **10000 bps**

Q17 Text Solution:

The Hamming distance between two binary strings is the number of positions at which the corresponding bits differ.

Compare the corresponding bits:

Position	String 1 (10101)	String 2 (11110)	XOR
1st bit	1	1	0
2nd bit	0	1	1
3rd bit	1	1	0
4th bit	0	1	1
5th bit	1	0	1

Total number of differing positions (no. of 1's) = 3.

The **Hamming distance** between 10101 and 11110 is **3**.

Q18 Text Solution:

Co mp on ent	De scri pti on	Ac ces s Ti me	Or de r (In cre asi ng Ac ces s Ti me)
E. Re gis ter s	Loc ate d in CP U, pro vid es fas t acc ess	1- 2 na no se pro co nd s (n s)	1st (Fa ste nd st)



D. SRAM	Used for CPU cache, fast but not as fast as registers	5-10 nanoseconds (ns)	2n
C. DRAM	Main memory in computer, slowest than SRAM	50-100 nanoseconds (ns)	3r
A. SSD	Solid-state storage device	100-200 nanoseconds (ns)	4th

	Access time (μs)	Cost
B. Optical Disks	100 to 1000 (ms)	5th (Slowest)

Correct Order (Increasing Access Time):

E (Registers) < D (SRAM) < C (DRAM) < A (SSD) < B (Optical Disks)

Q19 Text Solution:

The **Pumping Lemma for Regular Languages** is a property used to **prove that a language is not regular** by showing that no valid decomposition satisfies its conditions.

Q20 Text Solution:

Given Boolean Function:

$$F(A, B, C) = \sum m(0, 1, 3, 7)$$

This means the function is represented as a sum of minterms for the indices 0, 1, 3, and 7.

3-variable Karnaugh Map

A \ BC	00	01	11	10
0 (A')	1 (m0)	1 (m1)	1 (m3)	0 (m2)
1 (A)	0 (m4)	0 (m5)	1 (m7)	0 (m6)

Simplified Expression:

$$F(A, B, C) = A'B' + BC$$

Q21 Text Solution:

2-D viewing transformation pipeline:



C. Construct world-coordinates scene: First, create the scene in **world coordinates** using modeling transformations (positioning objects in the scene).

A. Convert world-coordinates to viewing coordinates: Transform the world coordinates into the **viewing coordinates**, considering the observer's viewpoint.

B. Map to normalized coordinates: Map the viewing coordinates to **normalized coordinates** using the window-viewpoint specifications.

D. Map to device coordinates: Finally, map the **normalized coordinates** to the **device coordinates**, fitting them to the screen or display.

Correct Order is :C → A → B → D

Q22 Text Solution:

We need to find the probability that **at least one** of them hits the target, i.e.,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Compute $P(A \cap B)$: Since A and B are independent shooters, the probability that **both** A and B hit the target is:

$$P(A \cap B) = P(A) \times P(B) = 1/4 \times 2/5 = 2/20 = 1/10$$

Compute $P(A \cup B)$:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 1/4 + 2/5 - 1/10$$

$$P(A \cup B) = \frac{5+8-2}{20}$$

$$P(A \cup B) = \frac{11}{20}$$

Final Answer: 11 / 20 (Option 4)

Q23 Text Solution:

A. Internet Message Access Protocol (IMAP) : IMAP allows email clients to **access email on the server** without downloading them.

B. Address Resolution Protocol (ARP) : ARP maps **IP addresses to MAC addresses** on a local network.

C. Post Office Protocol (POP) : POP is used by email clients to **retrieve emails from a mail server**, typically downloading them.

D. TELNET : TELNET is used for **remote login** to another computer over a network.

Correct Match : Option C: A-IV, B-III, C-I, D-II

Q24 Text Solution:

A **Reservation Table** is used to manage the scheduling and allocation of resources in dynamic systems, especially when dealing with **nonlinear patterns** and complex resource dependencies. When the pipeline follows a **nonlinear pattern**, this table becomes crucial for tracking resource allocation, timing, and conflicts, making it **more interesting** and dynamic.

Q25 Text Solution:

The **phantom problem** happens when **records are inserted** by other transactions while a query is being executed. This causes the query results to **change unexpectedly** and leads to **inconsistent reads** within the transaction.

Q26 Text Solution:

A. True. An NP-complete problem can be solved in polynomial time **if and only if** all NP-complete problems can be solved in polynomial time.

B. True. All NP-complete problems are a subset of NP-hard problems.

C. True. If an NP-hard problem can be solved in polynomial time, then all NP-complete problems can be solved in polynomial time.

D. True. Not all NP-hard problems are NP-complete because some NP-hard problems may not be in NP (e.g., non-decision problems).

Q27 Text Solution:

The disk scheduling graph corresponds to the **C-LOOK** algorithm. In the **C-LOOK** disk scheduling algorithm:

- The disk arm moves in one direction (either towards the higher or lower track numbers).
- Once it reaches the **farthest request** in that direction, instead of going to the end of the disk, it **jumps back** to the closest request in the opposite direction.
- It then continues servicing the requests in that direction.

Q28 Text Solution:

Observability: Helps you see the software's behavior, making it easier to test. (**Leads to**



testable software)

Controllability: Allows you to control the software's behavior during testing. (Leads to testable software)

Repairability: Refers to fixing defects, but doesn't affect how easily you can test the software. (Does not directly lead to testable software)

Understandability: Helps in designing and understanding test cases. (Leads to testable software)

The characteristic that **does not lead to testable software** is: **Option C: Repairability**

Q29 Text Solution:

The round-trip propagation time T_p is:

$$T_p = 2 \times \text{Propagation time}$$

$$T_p = 2 \times 25\mu s = 50\mu s$$

For CSMA/CD, the frame transmission time must be greater than or equal to the round-trip propagation time. Thus:

$$\text{Frame transmission time} \geq T_p = 50\mu s$$

The **minimum frame size** can be calculated by:

Frame size = Bandwidth $\times$ Frame transmission time

Since the bandwidth is 20Mbps = 20×10^6 bits per second, and the transmission time is $50\mu s = 50 \times 10^{-6}$ seconds:

$$\text{Frame size} = 20 \times 10^6 \times 50 \times 10^{-6}$$

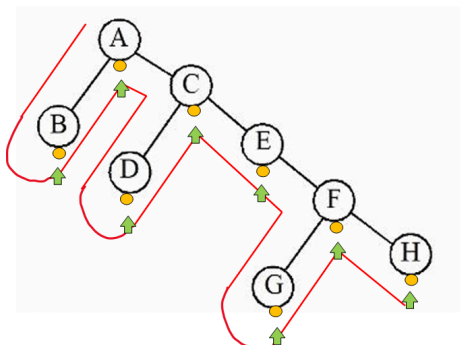
$$\text{Frame size} = 20 \times 50 = 1000 \text{ bits}$$

Q30 Text Solution:

To determine the in-order traversal of the given binary tree, we use the following traversal rule:

Traverse the **left subtree**, Visit the **root** node,

Traverse the **right subtree**.



In-order Traversal: B,A,D,C,E,G,F,H

Q31 Text Solution:

C. Redundant Instruction Elimination: Start by removing unnecessary or redundant instructions that don't impact the program's output. This makes the code more efficient right away.

D. Flow of Control Optimization: Next, focus on improving the control flow, like loops and conditional statements. This helps reduce the number of unnecessary checks or jumps in the program.

A. Algebraic Simplification: After that, simplify complex mathematical or logical expressions, making the code clearer and easier for the compiler to optimize.

B. Use of Machine Idioms: Then, replace high-level operations with low-level machine-specific instructions that execute faster on the target machine.

E. Improved Target Code: Finally, generate the optimized machine code, which reflects all the improvements made in earlier stages.

Q32 Text Solution:

B. Sender sends a window of frames: The sender starts by sending a window of frames to the receiver.

C. Receiver acknowledges the received frames: The receiver acknowledges the frames it has successfully received.

A. Sender moves the window to the next set of frames: Once the acknowledgment is received, the sender moves the window to send the next set of frames.

D. If acknowledgment is not received, sender retransmits the frame: If the acknowledgment is not received within a specified time, the sender retransmits the frame.

Q33 Text Solution:

A Deque (Double-Ended Queue) allows adding and removing elements at both ends (front and rear). It supports increment operations for adding or removing elements at either end, which is a key feature of a deque.



Q34 Text Solution:

A. This statement is **TRUE**. In a relation where all attributes are keys (meaning every attribute is part of a candidate key), there are no functional dependencies (FDs) that violate BCNF, so the relation is trivially in **BCNF**.

B. This statement is **TRUE**. If a relation violates **4NF** due to nontrivial **Multi-Valued Dependencies (MVDs)**, it must be decomposed to eliminate the violation and achieve **4NF**.

C. This statement is **TRUE**. Decomposition based on **MVDs** removes redundancy by ensuring that **nontrivial MVDs** are eliminated, which typically reduces data duplication.

D. This statement is **FALSE**. BCNF is stronger than 3NF. **BCNF** is a **stricter form** of normalization compared to **3NF** because every determinant in BCNF must be a candidate key, while 3NF allows certain exceptions, such as transitive dependencies where the non-prime attribute is dependent on another non-prime attribute. Hence, the false statement is **D** only.

Q35 Text Solution:

INSERT: Used to add new records to the table.

DELETE: Used to remove records from the table.

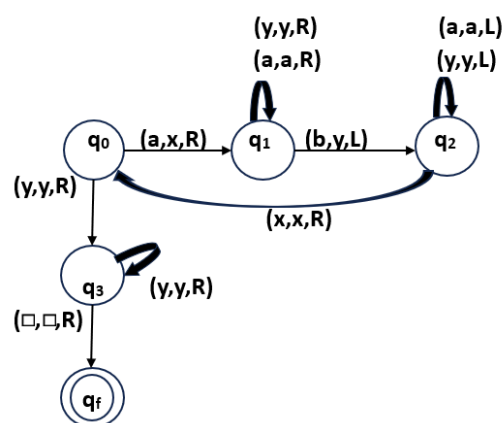
UPDATE: Used to modify existing records in the table.

SELECT: Used to retrieve data (does not modify data)

INSERT, DELETE, and UPDATE are used to modify the data.

Q36 Text Solution:

A **stack frame** refers to the entire set of parameters, local variables, and return address that are stored in the stack when a procedure or function is invoked. Each function call creates its own stack frame, and it is removed once the function returns.

Q37 Text Solution:**Analysis of the Language Accepted:**

This Turing machine is designed to match **a's** with **b's**, replacing them with **x** and **y**, respectively. It stops when all **a's** and **b's** have been paired, meaning the string is of the form **aⁿ bⁿ**.

- The machine accepts strings like **aⁿ bⁿ** where **n ≥ 1**.
- The machine moves through various states and processes pairs of **a** and **b** until the entire string has been processed.

Q38 Text Solution:

B. Fetch the instruction: The instruction is fetched from memory.

D. Decode the instruction: The fetched instruction is decoded to determine the operation to be performed and the operands involved.

A. Read the effective address: If the instruction involves memory access, the effective address of the operand(s) is calculated or read. This may involve further calculations or memory accesses, depending on the addressing mode.

C. Execute the instruction: The actual operation specified by the instruction is performed using the operands obtained.

Q39 Text Solution:

B: Spiral model is a risk-driven process model – This is correct. The Spiral model emphasizes risk analysis and management at every phase, which is a central part of the model.

D: Spiral model involves a set of anchor point milestones – This is also correct. In the Spiral model, key milestones (anchor points) are



identified at the end of each cycle to assess progress and make decisions about the next phase.

The other statements are incorrect because:

A: The Spiral model is not strictly an incremental model. It combines elements of both iterative and incremental models, but its primary focus is on risk management and refining the system.

C: The Spiral model is not an acyclic balancing approach; it is cyclic, as it involves multiple iterative cycles (spirals) throughout the project. Thus, the correct answer is **B & D Only**.

Q40 Text Solution:

We will determine the inverse of $x^2yx^4y^2$ in the given group G , where $x^5 = e$ and $y^3 = e$.

Step 1: Understanding the Given Information

- $x^5 = e \Rightarrow x^{-1} = x^4$ (since multiplying both sides by x^{-1} gives $x^4 = x^{-1}$)
- $y^3 = e \Rightarrow y^{-1} = y^2$ (since multiplying both sides by y^{-1} gives $y^2 = y^{-1}$)

Step 2: Finding the Inverse

To find the inverse of $x^2yx^4y^2$, we need to find an element g^{-1} such that:

$$(x^2yx^4y^2)(g^{-1}) = e$$

Using the property that the inverse reverses the order, the inverse of $x^2yx^4y^2$ is:

$$(y^2)^{-1}(x^4)^{-1}(y^{-1})(x^2)^{-1}$$

Using $y^{-1} = y^2$ and $x^{-1} = x^4$, we substitute:

$$(y^2)^{-1} = y, (x^4)^{-1} = x, (y^{-1}) = y^2,$$

$$(x^2)^{-1} = x^3$$

Thus, the inverse becomes: yx^3y^2x

Thus, the correct answer is: **Option B: yx^3y^2x**

Q41 Text Solution:

RA ID Level	Correct Match	Description
RA ID 1	A → II	Disk mirroring (data is duplicated on two disks for redundancy)

RA ID Level	Correct Match	Description
RA ID 2	B → IV	ECC (Error Correction Code) organization (data with error correction codes)
RA ID 3	C → I	Bit-interleaved parity organization (parity is distributed at the bit level across disks)
RA ID 4	D → III	Block-interleaved parity organization (parity is stored in blocks)

Q42 Text Solution:

A: Training time increases with the size of the network (**correct**).

B: While neural networks **can** run on conventional computers, their training and execution become highly inefficient for large-scale models. Training deep neural networks typically requires **GPUs or TPUs** for feasible performance. A conventional CPU may not handle complex models effectively, making the statement **incorrect**.

C: Neural networks are inspired by the brain and they mimic the same way as that of the humans brain (**correct**).

D: A neural network (like RNNs) include feedback (**correct**).

Q43 Text Solution:

LIST-I (Concepts)	LIST-II (Matching Terms)	Explanation
A. Conf	II. Prece	In parsing, precedence is



LIST-I (Concepts)	LIST-II (Matching Terms)	Explanation
Conflict Resolution	Dependency	used to resolve shift-reduce and reduce-reduce conflicts.
B. Common Subsequence	IV. Directed Acyclic Graph (DAG)	DAGs help eliminate common subexpressions, optimizing code by avoiding redundant calculations.
C. Quadruples	I. Three-Address Code	Quadruples are a representation format of three-address code (TAC) in intermediate code generation.
D. L-Attributed Definitions	III. Syntax-Directed Translations (SDTs)	L-Attributed Definitions (LADs) allow syntax-directed translations, where attributes are evaluated in a left-to-right manner.

Q44 Text Solution:

B (Choose task domain) – First, decide what real-world domain the knowledge base will represent.

C (Select atoms) – Define how to represent knowledge using atomic propositions.

D (Axiomatizing the domain) – Provide facts and rules that describe the true propositions.

A (Ask questions) – Validate the knowledge base by querying and refining it.

Q45 Text Solution:

The correct mathematical notation for **logical entailment** is:

$$\alpha \models \beta$$

which means " **α logically entails β** ", i.e., if **α is true, then β must also be true** in all models where α holds.

Q46 Text Solution:

LIST-I (Concepts)	LIST-II (Matching Terms)	Explanation
A. Breadth-First Search (BFS)	IV. Queue	BFS explores all nodes level by level using a queue (FIFO structure).
B. Depth-First Search (DFS)	III. Stack	DFS explores as deep as possible before backtracking using a stack (LIFO structure).
C. Prefix	I. LISP	LISP (List Processing Language)



LIST-I (Concepts)	LIST-II (Matching Terms)	Explanation
notation		extensively uses prefix notation for expressions.
D. Infix notation	II. Syntax Tree	Infix expressions are often represented using a syntax tree in parsing.

Q47 Text Solution:

Number of frames in physical memory is given by:

$$= \frac{\text{Number of frames}}{\text{Physical Memory Size}} = \frac{\text{Page Size}}{\text{Page Size}}$$

$$= \frac{4 \text{ MB}}{2 \text{ KB}}$$

Convert MB to KB:

$$4 \text{ MB} = 4 \times 1024 \text{ KB} = 4096 \text{ KB}$$

Now, divide:

$$\frac{4096 \text{ KB}}{2 \text{ KB}} = 2048$$

Q48 Text Solution:

Collaborative document inspection involves stakeholders working together to review and validate the requirements of the software. This technique helps ensure that the requirements are correct, complete, and align with user needs.

It is a part of the **Requirement Validation Process**, where the goal is to **validate** the requirements by checking them against user expectations and business goals before development begins.

Q49 Text Solution:

The basic syntactic elements of **First-Order Logic (FOL)** are:

- **A. Constant:** Represents specific, fixed objects in the domain.
- **C. Predicate:** Represents relations or properties of objects.
- **E. Function:** Defines mappings from one object in the domain to another.

Domain (B) refers to the set of all possible objects but is **not a syntactic element** itself—it's more of a semantic concept.

Temporal (D) is part of **temporal logic**, which is an extension of first-order logic and not a basic syntactic element.

Q50 Text Solution:

1. **Initial Values:** $x=10, y=15$

2. **Condition Check:**

- The conditional (ternary) operator evaluates the condition: $x==y$.
- Here, $10==15$ is false.

3. Since the condition is **false**, the false branch of the ternary operator is executed: $x = y - x$
 $x = 15 - 10 = 5$;

Final Value: After executing the statements, the value of x is **5**.

Q51 Text Solution:

Scan conversion is the process of converting geometric data (lines, shapes) into pixel data on a raster display.

This process can introduce visual artifacts due to the discrete nature of pixels on the screen.

A. Staircase Appearance: Also known as aliasing. It occurs when diagonal or curved lines are represented as jagged steps because pixels are square and discrete.



B. Unequal Brightness of Slanted Lines: The intensity of slanted lines may vary, causing uneven brightness.

C. Picket Fence Problem: Refers to the loss of thin lines or objects when their resolution aligns poorly with the pixel grid.

D. Rasterization: Rasterization is the process itself and not a side effect, so it is **not an adverse effect**.

E. Pre-filtering and Post-filtering: These are solutions to improve scan conversion, not side effects.
The correct combination of adverse side effects is: **A, B, and C only**.

Q52 Text Solution:

A **firewall** is a network security device or software that **monitors and controls** incoming and outgoing network traffic based on predefined security rules. It acts as a **barrier** between a trusted internal network and untrusted external networks (such as the internet), **filtering data packets** to allow or block **traffic** based on **security policies**.

Q53 Text Solution:

A. fseek():

- **Use III: fseek()** sets the position of the file pointer to a specific location in the file. This location is specified relative to a reference point (beginning, current position, or end of the file).

B. ftell():

- **Use I: ftell()** returns the current position of the file pointer, which indicates the byte offset from the beginning of the file. It helps you determine where you are in the file.
- For example: After reading a few bytes, you can use ftell() to find the current position in the file.

C. feof():

- **Use IV: feof()** checks whether the end of the file has been reached. It returns a non-zero

value if the file pointer is at the end of the file, otherwise it returns 0.

- It is often used in loops to read until the end of the file is reached.

D. rewind():

- **Use II: rewind()** resets the file pointer to the beginning of the file. It is equivalent to calling fseek(file_pointer, 0, SEEK_SET), but is specifically used for that purpose.
- It's often used to re-read a file from the start.
- Example: After reading a file and processing its contents, you can use rewind(file_pointer) to set the pointer back to the beginning.

Thus, the correct matching is **A-III, B-I, C-IV, D-II**.

Q54 Text Solution:

In a **data warehouse**, the **fact table** stores the **primary information**, which consists of **measurable business data** (e.g., sales, revenue, quantity sold). It references **dimension tables** that provide descriptive context.

- **Fact Table** → Stores numeric values (measurable data) and foreign keys to dimension tables.
- **Dimension Table** → Stores descriptive attributes (e.g., product name, customer details).
- **Lookup Table** → Used for mapping or validation.
- **Primary Table** → Not a standard term in data warehousing.

Q55 Text Solution:

The **INSERT** command in SQL is used to add data into a **relation (table)**. It can perform the following:

A. Add a single tuple (row) to a relation → Correct

- Example: INSERT INTO Students (ID, Name, Age) VALUES (101, 'Alice', 22);
- This adds a **single row** to the table.

B. Add multiple tuples (rows) to a relation → Correct



- Example: INSERT INTO Students (ID, Name, Age) VALUES (102, 'Bob', 23), (103, 'Charlie', 21);
- This inserts **multiple rows** at once.

C. Add values to specific attributes → Correct

- You can insert data into **specific columns**, leaving others as NULL or default.
- Example: INSERT INTO Students (ID, Name) VALUES (104, 'David');
- Here, only **ID** and **Name** are inserted, while **Age** remains NULL (if not set to default).

D. Insert a new table → Incorrect

- The **INSERT** command does **not** create a new table.
- The **CREATE TABLE** command is used to define a new table.

Q56 Text Solution:

The classification of grammars follows **Chomsky's Hierarchy**, which consists of **four types of grammars**:

Type	Name	Production Rule Form	Corresponding Match
Type-0	Recursively Enumerable Grammar	No restrictions on production rules	A - IV
Type-1	Context-Sensitive Grammar	Rules are of the form $x \rightarrow y$, where $ x < y $ and $x, y \in \{V \cup T\}^*$	B - III
Type-2	Context-Free	All production rules are of the	C - I

Type	Name	Production Rule Form	Corresponding Match
	Grammar (CFG)	form $A \rightarrow \alpha$, where A is a non-terminal, and α is a string of terminals & non-terminals. Includes Linear Grammar , where each production has at most one non-terminal .	
Type-3	Regular Grammar	Either Right Linear or Left Linear	D - II (GNF: Greibach Normal Form, related to regular grammars)

Q57 Text Solution:

Requirement management ensures that all software requirements are documented, tracked,



and linked properly. **Traceability tables** help maintain these links.

1. **Feature Traceability Table** → Tracks implementation of features.
2. **Dependency Traceability Table** → Manages requirement dependencies.
3. **Interface Traceability Table** → Tracks system interfaces.
4. **Page Traceability Table** → **Not part of requirement management** (not a recognized traceability table in software engineering).

Thus, **Option D is correct** because "**Page Traceability Table**" is **NOT** a part of requirement management.

Q58 Text Solution:

Graph	Formula	Example: $n = 4$	N u m b e r o f E d g e s
C_n	Edges = n	$E = 4$	4
W_n	Edges = $2n$	$E = 2(4)$	8
K_n	Edges = $\frac{n(n-1)}{2}$	$E = \frac{4(4-1)}{2}$	6
$K_{n,n}$	Edges = n^2	$E = 4^2$	16
Q_n	Edges = $n \cdot 2^{n-1}$	$E = 4 \cdot 2^3$	32

Thus, the general increasing order of edges (for $n > 3$) is:

$$C_n < W_n < K_n < K_{n,n} < Q_n$$

Q59 Text Solution:

We are given the Boolean function:

$$F = A'B + CD' + A'B + CD'$$

According to **De Morgan's Law**, the complement of a sum is the product of the complements:

Now, let's find the complement of each term:

$$1. A'B = A+B'$$

$$2. CD' = C'+D$$

Thus, the complement of $F = A'B + CD' + A'B + CD'$ is:

$$F' = (A+B') \cdot (C'+D) \cdot (A+B') \cdot (C'+D)$$

$$F' = (A+B') \cdot (C'+D)$$

Q60 Text Solution:

A **service routine** (or Interrupt Service Routine - ISR) is executed when an interrupt occurs. The correct sequence ensures proper handling of the interrupt while preserving the CPU state.

Step-by-Step Execution of Service Routine:

(A) Save contents of processor registers: Before handling the interrupt, the CPU registers are saved to prevent data loss.

(D) Check which flag is set: Identify the interrupt source by checking the status flags.

(C) Service the device whose flag is set: Execute the necessary operation to handle the interrupt (e.g., read input, write output).

(E) Restore contents of processor registers: Restore previously saved register values to resume execution correctly.

(B) Turn the interrupt facility on: Re-enable interrupts to allow new interrupts after the current one is serviced.

Thus, the correct order is **A → D → C → E → B**.

Q61 Text Solution:

A. Hadoop Distributed File System (HDFS) → III. Name-Node

- HDFS uses the Name-Node to store and manage the metadata and file system namespace.

B. Veracity → I. Trust



- Veracity is related to the trustworthiness and quality of data, making Trust the correct match.

C. Apache Pig → IV. Yahoo

- Apache Pig was originally developed by Yahoo for simplifying data processing on Hadoop.

D. Apache Hive → II. Facebook

- Apache Hive was developed at Facebook to provide SQL-like query capabilities for big data.

Thus, the correct match is **C. A-III, B-I, C-IV, D-II.**

Q62 Text Solution:

Convert inequalities to equations

1. $3x + 4y = 60$

- If $x = 0$, then $4y = 60 \Rightarrow y = 15 \rightarrow$ **Point (0,15)**
- If $y = 0$, then $3x = 60 \Rightarrow x = 20 \rightarrow$ **Point (20,0)**

2. $5x - 3y = 20$

- If $x = 0$, then $-3y = 20 \Rightarrow y = -\frac{20}{3}$
(Not valid as $y \geq 0$)
- If $y = 0$, then $5x = 20 \Rightarrow x = 4 \rightarrow$ **Point (4,0)**
- If $x = 7$, then
 $5(7) - 3y = 20 \Rightarrow 35 - 3y = 20 \Rightarrow y = 5$
 $\rightarrow$ **Point (7,5)**

Find the Optimal Solution

Evaluate $Z = 30x - 18y$ at feasible points:

Po int	$Z = 30x - 18y$
(4, 0)	$30(4) - 18(0) = 120$
(7, 5)	$30(7) - 18(5) = 210 - 90 = 120$
(0, 15)	$30(0) - 18(15) = -270$ (not feasible as we minimize Z)

Correct Answer: Option A (4,0) and (7,5) only

Q63 Text Solution:

Heap Sort does not use the divide and conquer method. It uses a **selection-based approach** by building a heap and sorting by extracting elements.

Q64 Text Solution:

A. True: Static members are shared across all instances of the class.

B. True: Friend functions can be called like normal functions without an object.

C. False: The scope resolution operator (::) cannot be overloaded.

D. True: Virtual inheritance solves duplication issues in **multiple inheritance**.

Q65 Text Solution:

The **degree** of a relation in a database refers to the **number of attributes (columns)** present in the relation (table). For example, if the Student relation has 4 attributes (Student_ID, Name, Age, Course), then the degree of the relation is **4**.

Q66 Text Solution:

$X \rightarrow RT/ TR$: This is in CNF because the right-hand side has exactly two non-terminals.

$T \rightarrow t$: This is in CNF because the right-hand side has exactly one terminal t .

$R \rightarrow XT/ r$: This is in CNF because the right-hand side has exactly two non-terminal XY or one terminal r .

All the productions either follow the form $A \rightarrow BC$ (two non-terminals) or $A \rightarrow a$ (one terminal), satisfying the CNF rules. Therefore, **Option B is in CNF.**

Q67 Text Solution:

Inorder traversal of a binary tree follows the sequence:

- D:** Start traversing by visiting the nodes in the rooted tree (This is the first step, as we must initiate traversal).
- A:** Visit the Left Subtree (As per inorder, we visit the left subtree first).
- B:** Visit the Root Node (After left, we visit the root).
- C:** Visit the Right Subtree (After root, we visit the right subtree).



- **E:** Repeat the above three steps (This ensures traversal continues recursively).

Thus, the correct order is: **D → A → B → C → E**

Q68 Text Solution:

LIST-I (Concepts)	LIST-II (Definitions)
A. Service Level Agreement (SLA)	III - A contract defining service performance, availability, and reliability.
B. Virtualization	IV - Allows multiple OS to run on a single physical server using virtual machines.
C. Internet of Things (IoT)	I - Connects and enables communication between physical devices via the Internet.
D. Platform as a Service (PaaS)	II - Provides a cloud-based environment to develop and deploy applications.

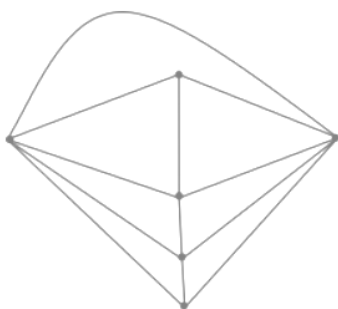
Q69 Text Solution:

A graph is said to be planar if it can be drawn in a plane so that no edge cross.

Figure A can be redraw as:



Figure B can be redraw as:



But, **Figure C** cannot be redrawn in such a way that no edges intersect each other.

Therefore, Figures A and B both are planar graphs.

Q70 Text Solution:

A. In connection-oriented service, the destination address is to be specified only during the setup.

- **Correct:** In a connection-oriented service (e.g., TCP), the destination address is specified during the connection setup phase, and subsequent packets use the established connection.

B. Packet sequencing is not guaranteed in connection-oriented service.

- **Incorrect:** In a connection-oriented service like TCP, packet sequencing is **guaranteed** because it ensures reliable data delivery in the correct order.

C. Flooding is a method in which every incoming packet is sent out on every outgoing line except the one by which it arrived.

- **Correct:** Flooding is a technique where a packet is forwarded on all outgoing links (except the incoming link) to ensure it reaches its destination.

D. Presentation layer performs detection and recovery from errors in the transmitted data.

- **Incorrect:** Error detection and recovery are performed by the **Data Link Layer** or the **Transport Layer**, not the Presentation Layer. The Presentation Layer handles data translation, encryption, and compression.

Correct Answer: Option C: A & C only

Q71 Text Solution:

Prim's Algorithm: This is a greedy algorithm used to find the Minimum Spanning Tree (MST) of a graph, and it uses **BFS** in its process, particularly when selecting the next vertex to be added to the MST.

Dijkstra's Algorithm: This is a shortest path algorithm that, when implemented using a priority queue, can be seen as a **BFS** variant since



it explores nodes layer by layer to find the shortest path.

Q72 Text Solution:

We can solve this recurrence using the **Master Theorem**. The recurrence is of the form:

$$T(n) = T\left(\frac{n}{b}\right) + f(n)$$

where:

- $a=1$ (because we only have one subproblem),
- $b = \frac{3}{2}$ (since $\frac{n}{b} = \frac{2n}{3}$, so $b = \frac{3}{2}$)
- $f(n)=1$ (since the additional work done is constant).

Now, let's apply the Master Theorem:

We need to compare $f(n)$ with $n^{\log_b a}$, where:

$$\log_b a = \log_{\frac{3}{2}} 1 = 0$$

$$\text{So, } n^{\log_b a} = n^0 = 1$$

Since $f(n)=1$, we are in the case where $f(n) = \Theta(n^0)$, which matches the second case of the Master Theorem:

- If $f(n) = \Theta(n^{\log_b a})$, then the solution to the recurrence is $T(n) = \Theta(n^{\log_b a} \log n)$.

Thus, $T(n) = \Theta(1 \cdot \log n) = \theta(\log n)$.

The correct answer is **Option B: $\theta(\log n)$** .

Q73 Text Solution:

B) Domain: "Domain" refers to the context or framework for actions but isn't considered a primary component of the "classic planning definition". While **Init, Action, and Goal** are the standard components in a classical planning problem.

Q74 Text Solution:

Yacc (Yet Another Compiler Compiler) is a tool used to generate parsers for context-free grammars. The parser generated by Yacc is an **LALR (Look-Ahead LR)** parser, which is a type of **LR parser**.

- **LALR** parsers are a more memory-efficient version of **LR parsers** (which are powerful parsers capable of handling a wide range of context-free grammars).
- **Yacc** generates an LALR parser because it balances the efficiency of parsing with the

ability to handle complex grammars.

Q75 Text Solution:

The definition organized into **Thinking Humanly**, **Thinking Rationally**, **Acting Humanly**, and **Acting Rationally** refers to **Artificial Intelligence (AI)**. These categories describe different ways of approaching AI:

- **Thinking Humanly:** Simulating human-like cognition.
- **Thinking Rationally:** Reasoning and decision-making based on logic.
- **Acting Humanly:** Imitating human actions and behaviors.
- **Acting Rationally:** Acting in ways that maximize goal achievement efficiently.

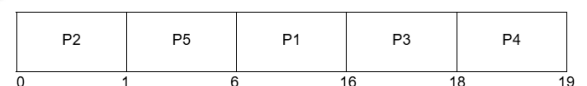
These four categories help to outline the scope and various methods of **AI** development.

Q76 Text Solution:

Scheduler: The **scheduler** does not directly **interrupt a running process**. Instead, it manages when processes are executed and assigns CPU time. It is a part of the operating system responsible for **process management**, but it doesn't cause interruptions; it determines which process should run next based on scheduling algorithms.

Q77 Text Solution:

Gantt Chart:



Calculate Average Waiting Time :

P					T
r			P		A
			r		W
			i		T
					T
					=
					(
					(T
					C
					A
					T
					T
					-
					-
					A
					B
					T
					T
					)



P1	0	1	3	1	1	6
P2	0	1	1	1	1	0
P3	0	2	4	1	1	1
P4	0	1	5	1	1	1
P5	0	5	2	6	6	1

$$\text{Average Waiting Time} = \frac{6+0+16+18+1}{5} = 8.2$$

Q78 Text Solution:

LIST-I	LIST-II	Explanation
A. Project Join Normal Form	II. Fifth Normal Form	Project Join Normal Form (PJNF) is related to Fifth Normal Form (5NF) , as PJNF is a decomposition of a relation into smaller relations that ensures it satisfies 5NF.
B. The lost update problem	I. Concurrent system	The lost update problem occurs in a concurrent system when multiple transactions

LIST-I	LIST-II	Explanation
		overwrite each other's updates, leading to data inconsistency.
C. Discretionary Access Control	IV. Security	Discretionary Access Control (DAC) is a security mechanism where the owner of the resource decides the access permissions.
D. Data Manipulation Language	III. SQL	Data Manipulation Language (DML) is part of SQL , and it includes commands like INSERT, UPDATE, and DELETE to manipulate data in databases.

Q79 Text Solution:



LIST-I (Boolean Algebra Law)	LIST-II (Axioms)	Explanation
A. Absorption Law	IV. $a + (a * b) = a$	The Absorption Law states that $a + (a * b) = a$. This simplifies the expression.
B. Bounded Law	I. $a + 1 = 1$	The Bounded Law states that $a + 1 = 1$, meaning any value OR'd with 1 gives 1.
C. Identity Law	II. $a + 0 = a$	The Identity Law states that $a + 0 = a$, meaning any value OR'd with 0 gives the same value.
D. Distributive Law	III. $a * (b + c) = (a * b) + (a * c)$	The Distributive Law expresses the distributive property: $a * (b + c) = (a * b) + (a * c)$.

Q80 Text Solution:

Fluorescence refers to the light emitted by a phosphor when it is exposed to an electron beam or other types of energy sources. This light emission occurs almost immediately after the exposure.

Q81 Text Solution:

A. Red-Black Tree : True. A Red-Black Tree is a balanced binary search tree that ensures the worst-case time complexity for search, insertion, and deletion is $O(\log n)$ due to its balancing properties.

B. Trie - True. A Trie is a tree-like structure used for **efficient prefix-based searches**. It's particularly useful in applications like auto-completion or dictionary lookups.

C. AVL Tree - True. An **AVL Tree** is a self-balancing binary search tree, and it has a stricter balancing criterion than other trees, ensuring that the heights of the left and right subtrees of any node differ by at most 1.

D. B-Tree - True. A B-Tree is optimized for storage systems like disks, where multiple keys are stored in each node. It ensures efficient search, insert, and delete operations by minimizing the number of disk accesses required.

Q82 Text Solution:

LIST-I	LIST-II	Explanation
A. $(100110011)_2$	IV. $(1133)_{16}$	$(100110011)_2$ is a binary number, and its corresponding match is a hexadecimal number $(133)_{16}$.
B. Complement	III. Negative numbers	Complements refers to representing negative numbers in binary using



LIS T-I	LI ST -II	Explanation
	m be r	methods like two's complement.
C. Par ity Bit s	I. Er ro r d et ec ti on co d e	Parity bits are used in error detection by adding extra bits to check data correctness during transmission.
D. Var iabl e len gth inst ruc tio n	II. Cl S C	Variable length instruction is a feature of CISC (Complex Instruction Set Computing) , where instructions can have varying lengths.

Q83 Text Solution:

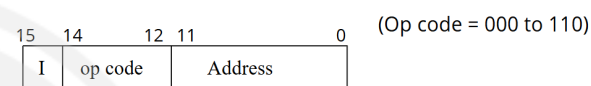
- **Call:** Call is a control flow instruction. It's used to jump to a subroutine or function. It doesn't directly manipulate data within registers or memory in the same way as other instructions.
- **Load** (Loud in the option): Load (or a similar instruction like Move) is a data transfer instruction. It moves data from one location (e.g., memory) to another (e.g., a register). While it moves data, it's primarily a transfer, not a manipulation.

- **And:** And is a logical instruction. It performs a bitwise AND operation on data, thus **manipulating the data**.
- **Increment:** Increment adds 1 to a value, directly **manipulating the data**.
- **Shift:** Shift moves the bits of a value left or right, thus **manipulating the data**.

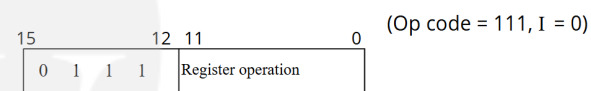
Therefore, **Call** and **Load** are not data manipulation instructions.

Q84 Text Solution:**Memory reference instructions**

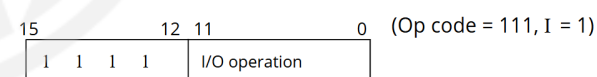
- Address part = 12 bits long.
- Addressing mode = 1 bit.
- Opcode = 3bits (from 000 to 110).

**Register reference instructions**

- Addressing mode (1 bit) = 0.
- Opcode (3bits) = 111.

**Input-output reference instructions**

- Addressing mode (1 bit) = 1.
- Opcode (3bits) = 111.



Hence, **Options B and C** do not represent memory reference instructions; they are **register reference instructions**.

Q85 Text Solution:

A. Boolean function, B. Continuous function, and C. Arbitrary function are all types of functions that feedforward networks with backpropagation can approximate.

D. Greedy function is unrelated because **greedy algorithms** focus on optimization strategies, not on backpropagation in neural networks.

Q86 Text Solution:

LIST-I (Algorithm)	LIST-II (Time Complexity)	Reason
A. Insertion Sort	III. $O(n^2)$ (Best Case)	When the array is already sorted, Insertion Sort runs in $O(n)$.
B. Binary Search	I. $O(\log n)$	Binary Search divides the search space in half at each step, resulting in $O(\log n)$ complexity.
C. Quick Sort	IV. $O(n \log n)$ (Average Case)	Quick Sort efficiently sorts data with an average-case time complexity of $O(n \log n)$.
D. Selection Sort	II. $O(n^2)$	Selection Sort always runs in $O(n^2)$ time, as it repeatedly selects the minimum element.

Q87 Text Solution:

The correct sequence of the Software Engineering Institute (SEI) Capability Maturity Model (CMM) stages is:

A. Initial: The process is ad-hoc and chaotic.

B. Repeatable: Basic project management processes are established to track cost, schedule, and functionality.

E. Defined: The process is well-defined and documented.

C. Managed: Detailed metrics are used to monitor the processes.

D. Optimizing: Continuous process improvement is implemented.

Q88 Text Solution:

The correct sequence of steps in the DHCP (Dynamic Host Configuration Protocol) process is:

B. DHCP DISCOVER: The client sends a DHCP DISCOVER message to find available DHCP servers.

C. DHCP OFFER: DHCP servers respond with a DHCP OFFER message, which includes an available IP address and other configuration details.

D. DHCP REQUEST: The client sends a DHCP REQUEST message, requesting the offered IP address from one of the servers.

A. DHCP ACK: The DHCP server responds with a DHCP ACK message, confirming the allocation of the IP address and finalizing the process.

E. Client receives IP configuration: The client receives the IP configuration, including IP address, subnet mask, and gateway.

Q89 Text Solution:

Statement C: $\emptyset \in \{\emptyset\}$ (True)

- The set $\{\emptyset\}$ contains one element, which is $\emptyset$.
- Since $\emptyset$ is explicitly listed as an element, this statement is **true**.

Statement E: $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ (True)

- The set $\{\emptyset\}$ contains one element: $\emptyset$.
- The set $\{\emptyset, \{\emptyset\}\}$ contains two elements: $\emptyset$ and $\{\emptyset\}$.
- Since $\{\emptyset\}$ is a subset (as all elements of $\{\emptyset\}$ are in $\{\emptyset, \{\emptyset\}\}$), this statement is **true**.

Q90 Text Solution:

A (True): Each module should follow the **Single Responsibility Principle (SRP)** and do only one thing.

B (True): Modules should communicate **only**



through a calling module (ensuring modularity).

C (False): Direct communication **between unrelated modules is not allowed** as it violates modularity.

D (True): Modern programming enforces **single-entry, single-exit** using structured control flow. Thus, the correct option is **B (A, B & D only)**.

Q91. Text Solution:

Starts and Ends with any combination: The initial state (1) has a self-loop for both 'a' and 'b', allowing any combination at the beginning. Crucially, the *final state* (4) *also* has a self-loop for both 'a' and 'b', allowing any combination at the end. This is essential.

"abb" as a substring: The path 1 → 2 → 3 → 4 represents the substring "abb".

Combining the Requirements: Option C correctly combines these. You can have any combination of 'a's and 'b's at the start (due to the self-loop on state 1), then the substring "abb" (moving through states 2, 3, and 4), and then *any combination of 'a's and 'b's at the end* (due to the self-loop on the final state 4).

Q92. Text Solution:

$$S \rightarrow AabbA$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

Here, A generates any string over $\{a, b\}$ (including the empty string). Thus, $S \rightarrow AabbA$ generates any string that can be split into some string x , followed by **abb** (as substring), followed by some string y . This exactly captures our language.

Q93. Text Solution:

All the given options exhibit the behavior of NFA, not DFA.

Q94. Text Solution:

Transitions for the Full DFA:

State	Input 'a'	Input 'b'	Explanation
q1	q2	q1	Start looking for "abb"; on 'a' go to q2, stay at q1 on 'b' (no "abb" found).

State	Input 'a'	Input 'b'	Explanation
q2	q2	q3	Stay in q2 on 'a'; on 'b', go to q3 (matching "ab").
q3	q2	q4	On 'a', go to q2 (reset to matching "a"); on 'b', go to q4 (found "abb").
q4	q5	q6	After "abb", go to q5 on 'a', and to q6 on 'b'. Accept strings that contain "abb".
q5	q5	q6	Stay in q5 on 'a'; on 'b', transition to q6. Accept any string after "abb".
q6	q5	q6	On 'a', transition to q5; stay in q6 on 'b'. Accept any string after "abb".

Final States:

- **q3, q4, and q5** are **accepting states**, because they all represent situations where "abb" has been matched, and the string can continue with any combination of 'a' and 'b' after that.

Q95. Text Solution:

$$(a+b)^*abb(a+b)^*$$

- This regular expression matches strings that:
 1. Start with **any combination of "a" and "b"** (denoted by $(a+b)^*$),
 2. Contain the substring **"abb"**,
 3. End with **any combination of "a" and "b"**.

Q96. Text Solution:

We are given a set of processes with **arrival times (AT)**, **burst times (BT)**, and **priorities**. Lower priority numbers indicate higher priority. We need to compute the **average response time (ART)** using the **Non-Preemptive Priority Scheduling Algorithm**.

Gantt Chart-

P2	P5	P1	P4	P3	
0	3	4	9	12	14

Compute Response Time (RT)

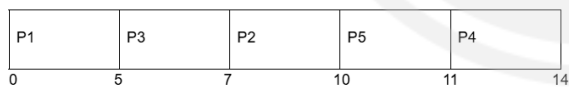


Process	BT	Priority	CT	TAT = CT - AT	WT = TAT - BT	Response Time (RT = WT)
P2	3	1	3	3 - 0 = 3	3 - 3 = 0	0
P5	1	2	4	4 - 0 = 4	4 - 1 = 3	3
P1	5	3	9	9 - 0 = 9	9 - 5 = 4	4
P4	3	4	12	12 - 0 = 12	12 - 3 = 9	9
P3	2	5	14	14 - 0 = 14	14 - 2 = 12	12

$$\text{Average Response Time} = \frac{\sum RT}{\text{Total Processes}} = \frac{0+3+4+9+12}{5} = \frac{28}{5} = 5.6 \text{ ms}$$

Q97. Text Solution:

Gantt Chart-



Compute Average Weighted TAT :

Process	AT	BT	CT	TAT = CT - AT	WT = TAT - BT	Weighted TAT (TAT / BT)
P1	0	5	5	5 - 0 = 5	5 - 5 = 0	1.0
P2	2	3	10	10 - 2 = 8	8 - 3 = 5	2.67

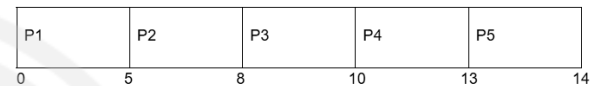
P3	2	2	7	5	3	2.5
P4	5	3	14	9	6	3.0
P5	6	1	11	5	4	5.0

$$\text{Average Weighted TAT} = \frac{1.0+2.67+2.5+3.0+5.0}{5} = \frac{14.17}{5} = 2.83$$

The average weighted turnaround time for the given processes using the HRRN scheduling algorithm is **2.83**.

Q98. Text Solution:

Gantt Chart:



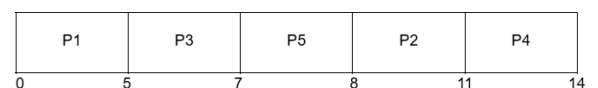
Compute Average Turnaround Time:

Process	AT	BT	CT	Turnaround Time (TAT) = CT - AT
P1	0	5	5	5 - 0 = 5
P2	2	3	8	8 - 2 = 6
P3	2	2	10	10 - 2 = 8
P4	5	3	13	13 - 5 = 8
P5	6	1	14	14 - 6 = 8

$$\text{Average Turnaround Time} = \frac{5+6+8+8+8}{5} = 7.0 \text{ ms}$$

Q99. Text Solution:

Gantt Chart:



Compute Average Waiting Time:

Process	AT	BT	CT	TAT = (CT-AT)	WT= (TAT-BT)
P1	0	5	5	5	0
P3	2	2	7	5	3
P2	2	3	11	9	6
P5	6	1	8	2	1
P4	5	3	14	9	6

$$\begin{aligned}
 \text{Average Waiting Time} &= \frac{\sum WT}{\text{Number of Processes}} \\
 &= \frac{0+3+6+1+6}{5} \\
 &= \frac{16}{5} = 3.2 \text{ ms}
 \end{aligned}$$

Q100. Text Solution:

HRRN is a non-preemptive scheduling algorithm that **reduces starvation** by prioritizing processes based on their **response ratio**.

It **dynamically adjusts priority** based on waiting time and burst time using the formula:

$$\text{Response Ratio} = \frac{\text{Waiting Time} + \text{Burst Time}}{\text{Burst Time}}$$

A process with a longer waiting time gets a higher priority, **preventing starvation** of long processes.



[Android App](#) | [iOS App](#) | [PW Website](#)